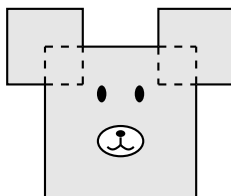


1. Ben the Bear's head is constructed using one square of side length 2 and two squares of side length 1, where all the squares have parallel sides, and the centers of the small squares are adjacent vertices of the large square. Find the perimeter around Ben's head.



Answer: 12

Solution:

Each of the small squares has two full sides that contribute to the perimeter and two half sides that contribute to the perimeter, so each small square contributes $1 + 1 + \frac{1}{2} + \frac{1}{2} = 3$ to the perimeter.

The large square has one full side, two $\frac{3}{4}$ sides, and one half side that contribute to the perimeter, so the large square contributes $2 + \frac{3}{2} + \frac{3}{2} + 1 = 6$ to the perimeter.

Thus, the perimeter of the entire shape is $3 + 3 + 6 = \boxed{12}$.

2. In 1 hour, Isaac can make 5 more burgers than Nathan can. Nathan can make 20 more burgers in 4 hours than Isaac can make in 3 hours. Together, how many burgers can Isaac and Nathan make in 5 hours?

Answer: 375

Solution: Let x be the number of burgers Nathan can make in an hour. Isaac can make $x + 5$ burgers in an hour, so $4x = 3(x + 5) + 20$. By the distributive property,

$$3(x + 5) + 20 = 3x + 15 + 20 = 3x + 35.$$

Thus, $4x = 3x + 35$. Subtracting $3x$ from both sides gives $x = 35$.

Together, Isaac and Nathan can make $35 + 40 = 75$ burgers in an hour, so they can make $75 \times 5 = \boxed{375}$ burgers in 5 hours.

3. The boxes in the expression below are filled with the numbers 1, 2, 3, 4, 5, and 6 so that each number is used exactly once and the value of the expression is **positive**. What is the least possible value of the expression?

$$\square \times \square \times \square - \square \times \square \times \square$$

Answer: 6

Solution: For all but one of the configurations, there is at least one even number in each product, so the value of the expression will be even. The only way it is not even is when all of the odd numbers are in one product and the even numbers are in the other, giving

$$2 \times 4 \times 6 - 1 \times 3 \times 5 = 48 - 15 = 33.$$

The only possible odd value is 33, but we can do a lot better.

Next, we should check whether a value of 2 or 4 is possible. For this to occur, we would need to put 3 and 6 in the same product. Otherwise both products would be even multiples of 3, so the sum would also be an even multiple of 3 and therefore a multiple of 6.

The closest cases are

$$2 \times 3 \times 6 - 1 \times 4 \times 5 = 36 - 20 = 16$$

and

$$2 \times 4 \times 5 - 1 \times 3 \times 6 = 40 - 18 = 22.$$

Now that we have confirmed 2 and 4 are not possible, we check if 6 is possible. Indeed, we have

$$2 \times 3 \times 5 - 1 \times 4 \times 6 = 30 - 24 = \boxed{6}.$$

4. Shrey wrote down six distinct positive factors of 30 and noticed that the average of these six numbers is 6. What is the largest number that Shrey wrote?

Answer: 15

Solution: The sum of all eight factors of 30 is $1 + 2 + 3 + 5 + 6 + 10 + 15 + 30 = 72$. Since the average of Shrey's numbers is 6, they must add up to $6 + 6 + 6 + 6 + 6 + 6 = 36$.

Therefore, the two numbers that Shrey did not write must add to $72 - 36 = 36$. This can only happen if he did not write 30 and 6. Thus, the largest number Shrey wrote is the next largest factor after 30, which is $\boxed{15}$.

5. Wen starts with a two-digit positive integer. She begins by adding the two digits together and subtracting 9 from that sum. Then, she cubes the result, adds 3, subtracts 22, and finally squares the number. If she ends up with a result of 2025, what is the smallest number she could have started with?

Answer: 49

Solution: Let s be the sum of the two starting digits. The final number is

$$((s - 9)^3 - 19)^2 = 2025,$$

so $(s - 9)^3 - 19 = 45$ or -45 . These give

$$(s - 9)^3 = 64 \quad \text{or} \quad (s - 9)^3 = -26.$$

Since s is an integer and -26 is not a perfect cube, only the first case has a valid solution. Thus $s = 13$. To minimize the two-digit number, we minimize its tens digit. Its ones digit is at most 9, so its tens digit is at least $13 - 9 = 4$. The answer is $\boxed{49}$.

6. Find the number of multiples of 15 between 100 and 999 that have a sum of digits strictly greater than 15.

Answer: 10

Solution: Note that all multiples of 15 have a units digit of 0 or 5. Then, because all multiples of 15 are multiples of 3, the sum of the digits must be a multiple of 3 that is larger than 15.

If a three-digit number ends in 0, then the largest sum of digits is $9 + 9 + 0 = 18$ in the case of 990. This is the only number ending in zero satisfying these conditions.

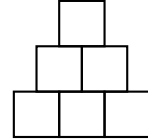
If a three-digit number ends in 5, then the largest sum of digits is $9 + 9 + 5 = 23$. Therefore, we care about the number three-digit numbers ending in 5 that have a sum of digits 18 or 21.

There are 6 numbers with a sum of 18: 495, 585, 675, 765, 855, 945, and there are 3 numbers with a sum of 21: 795, 885, 975.

The final answer is $1 + 6 + 3 = \boxed{10}$.

7. Kiran's coloring sheet has six squares, as shown below. Kiran will color each square a different color: red, orange, yellow, green, blue, or purple. How many different ways can the squares be colored according to the following rules?

- The red square touches the purple square and the orange square.
- The yellow square touches the orange square and the green square.
- The blue square touches the green square and the purple square.



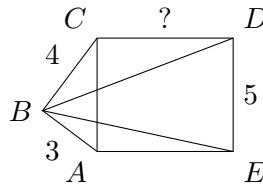
Answer: 12

Solution: The squares form a loop of adjacencies: red-orange-yellow-green-blue-purple-red. Each square must touch the two squares adjacent to it in this loop.

First, there are 6 ways to place the square on the top. Then, there are 2 ways to place the squares on the middle layer, since those must be the two squares adjacent to the top square. From there, the bottom layer is completely determined.

We conclude that there are $6 \cdot 2 = \boxed{12}$ ways to color the squares.

8. Triangle $\triangle ABC$ has a right angle at B and $AB = 3$, $BC = 4$. Rectangle $ACDE$ is constructed outside $\triangle ABC$. Given the area of triangle $\triangle BDE$ is $\frac{7}{12}$ of the area of pentagon $ABCDE$, find CD .



Answer: 6

Solution: Let $AE = CD = h$. The area of triangle $\triangle ABC$ is $\frac{1}{2} \cdot 3 \cdot 4 = 6$. Since the area of rectangle $ACDE$ is $5 \cdot h$, the area of pentagon $ABCDE$ is $6 + 5h$.

Let x be the length of the altitude from B to AC . The area of $\triangle ABC$ is $6 = \frac{1}{2} \cdot 5 \cdot x \implies x = \frac{12}{5}$. Treating $\overline{DE}$ as the base of triangle $\triangle BDE$, the height of $\triangle BDE$ is $\frac{12}{5} + h$. Thus, the area of $\triangle BDE$ is $\frac{1}{2} \cdot 5 \cdot (\frac{12}{5} + h) = 6 + \frac{5h}{2}$.

From the area ratio given in the problem statement,

$$\begin{aligned} \frac{6 + \frac{5h}{2}}{6 + 5h} &= \frac{7}{12} \\ \left(6 + \frac{5h}{2}\right) \cdot 12 &= (6 + 5h) \cdot 7 \\ 72 + 30h &= 42 + 35h \\ 30 &= 5h \\ \boxed{6} &= h. \end{aligned}$$

9. Recall that in an arithmetic sequence, the difference between consecutive terms is constant, and in a geometric sequence, the ratio of consecutive terms is constant. The common difference of the arithmetic sequence $\{a_n\}$ equals the common ratio of the geometric sequence $\{g_n\}$. Given $a_1 = 0$, $g_1 = 1$, and $a_1 + a_2 + a_3 = g_1 + g_2 + g_3$, find the unique positive integer x for which $a_x = g_x$.

Answer: 2

Solution: Let the common difference/ratio be r . Then we have that

$$\begin{aligned} 0 + r + 2r &= 1 + r + r^2 \\ 0 &= r^2 - 2r + 1 \\ &= (r - 1)^2 \end{aligned}$$

so the common difference/ratio is 1. Thus, $a_n = n - 1$ and $g_n = 1$ for all n . The common x th term is $a_x = x - 1 = 1 = g_x$, which gives $x = \boxed{2}$.

10. Arthur chooses three distinct integers between 2 and 14, inclusive, to be his winning numbers. If he rolls a fair 6-sided die and a fair 8-sided die with faces labeled 1–6 and 1–8, respectively, there is a $\frac{1}{3}$ chance that the sum of the faces rolled is a winning number. What is the largest winning number Arthur could have chosen?

Answer: 11

Solution: There are $6 \cdot 8 = 48$ possible outcomes of the dice, so we need to cover $48/3 = 16$ outcomes with out three numbers.

The most common sums are 7, 8, and 9, which can each be rolled in 6 ways. In order to cover 16 outcomes, the winning numbers must cover $5 + 5 + 6$ or $6 + 6 + 4$ outcomes. Since larger outcomes are less likely, the case $6 + 6 + 4$ will give the correct answer.

There is 1 way to get a sum of 14, 2 ways to get 13, 3 ways to get 12, and 4 ways to get $\boxed{11}$. If one of the winning numbers is 12 or more, we can only cover $3 + 6 + 6 = 15$ outcomes, which is not enough.

11. For each real number n , let $\lfloor n \rfloor$ denote the greatest integer less than or equal to n . Find the sum of the cubes of all distinct real numbers x satisfying the equation

$$x^3 - 3\lfloor x \rfloor = 4.$$

Answer: 15

Solution:

Let $k = \lfloor x \rfloor$, where k must be an integer. The equation can be rewritten as $x^3 = 3k + 4$, which implies $x = \sqrt[3]{3k + 4}$.

By the definition of the floor function, $k \leq x < k + 1$. Substituting the above expression for x yields

$$k \leq \sqrt[3]{3k + 4} < k + 1.$$

Since $f(t) = t^3$ is a strictly increasing function, we can cube all parts of the inequality without changing its direction:

$$k^3 \leq 3k + 4 < (k + 1)^3$$

This provides two conditions that the integer k must satisfy:

- (a) $k^3 \leq 3k + 4 \implies k^3 - 3k - 4 \leq 0$. By testing integer values, we find this inequality holds for $k \leq 2$.
- (b) $3k + 4 < (k + 1)^3 \implies 3k + 4 < k^3 + 3k^2 + 3k + 1 \implies 0 < k^3 + 3k^2 - 3$. By testing integer values, we find this inequality holds for integers $k \geq 1$ and for $k = -2$.

The integers k that satisfy both conditions are the intersection of the solution sets, which is $k \in \{-2, 1, 2\}$. We find the corresponding value of x for each valid k :

- If $k = -2$, then $x = \sqrt[3]{3(-2) + 4} = \sqrt[3]{-2}$.
- If $k = 1$, then $x = \sqrt[3]{3(1) + 4} = \sqrt[3]{7}$.
- If $k = 2$, then $x = \sqrt[3]{3(2) + 4} = \sqrt[3]{10}$.

Thus, the sum of the cubes of the solutions is $-2 + 7 + 10 = \boxed{15}$.

12. A right rectangular **pyramid** has height 4, and each of its lateral **edges** forms a 30° angle with the base. Given one of the four lateral faces has area $4\sqrt{15}$, what is the area of the pyramid's base?

Answer: $16\sqrt{11}$ or $\frac{176}{\sqrt{11}}$

Solution:

Let the base of the pyramid have sides of length $2x$ and $2y$. Then

$$\sqrt{x^2 + y^2} = h \cot 30^\circ = 4\sqrt{3}.$$

Without loss of generality, let $2y$ be the length of the base of the face with area $4\sqrt{15}$, then

$$y\sqrt{h^2 + x^2} = 4\sqrt{15}.$$

Substituting $x^2 = 48 - y^2$ gives

$$y\sqrt{16 + (48 - y^2)} = 4\sqrt{15} \implies y^2(64 - y^2) = 240.$$

Thus $y = 2$, $x = \sqrt{44}$, and $A = (2x)(2y) = \boxed{16\sqrt{11}}$.

13. For real numbers a and b with $a + b \neq 0$, define

$$a \otimes b = \frac{a}{a + b}.$$

The operation $\otimes$ is not *associative* since $(1 \otimes 1) \otimes 1 = \frac{1}{3}$ and $1 \otimes (1 \otimes 1) = \frac{2}{3}$ are different. Andrew evaluates $1 \otimes 1 \otimes 1 \otimes 1 \otimes 1 \otimes 1 \otimes 1$ in any order he likes. What is the absolute difference between the largest and smallest possible results that Andrew could obtain?

Answer: $\frac{5}{7}$

Solution: Smaller values of $a \otimes b$ are attained when a is very small and b is very large. We know that b can't be bigger than 1, since for positive numbers a and b , $a < a + b \implies a \otimes b < 1$. Thus, the largest possible value is $b = 1$.

We need the smallest possible value of $a = 1 \otimes 1 \otimes 1 \otimes 1 \otimes 1 \otimes 1$. This is the very similar to the problem we started with! Continuing the same reasoning, we find that the smallest possible

value of the expression is obtained by evaluating from left to right. This evaluates to $1 \otimes 1 = \frac{1}{2}$, $\frac{1}{2} \otimes 1 = \frac{1}{3}$, $\frac{1}{3} \otimes 1 = \frac{1}{4}$, $\frac{1}{4} \otimes 1 = \frac{1}{5}$, $\frac{1}{5} \otimes 1 = \frac{1}{6}$, $\frac{1}{6} \otimes 1 = \frac{1}{7}$.

By an analogous argument, the largest possible value of the expression is achieved by first evaluating the last six ones from left to right, then performing the remaining operation, which

yields $\frac{6}{7}$. The answer is $\frac{6}{7} - \frac{1}{7} = \boxed{\frac{5}{7}}$.

14. Find the sum of all real numbers x satisfying

$$\left(\sqrt{\frac{x^2 - 3x - 4}{6}} \right)^{x^2 + 3x - 18} = 1.$$

Recall that $(\sqrt{-4})^2 = -4$.

Answer: 2

Solution: Let $a(x) = \sqrt{\frac{x^2 - 3x - 4}{6}}$ and $b(x) = x^2 + 3x - 18$. There are three cases that could result in $a(x)^{b(x)} = 1$:

- $a(x) = 1$
- $a(x) \neq 0$ and $b(x) = 0$
- $a(x) = i$ and $b(x) = 4k$, where k is an integer

For the first case, note that $a(x) = \sqrt{\frac{x^2 - 3x - 4}{6}} = 1$ only occurs if $\frac{x^2 - 3x - 4}{6} = 1$. Solving this yields the equation $x^2 - 3x - 10 = (x - 5)(x + 2) = 0$, which has solutions $x = -2$ and $x = 5$.

For the second case, factor the quadratics present in both expressions to obtain $a(x) = \sqrt{\frac{(x+1)(x-4)}{6}}$ and $b(x) = (x+6)(x-3)$. Thus, $b(x) = 0$ at $x = -6$ and $x = 3$, and $a(x) \neq 0$ at both of these values of x .

For the third case, note $a(x) = i$ if and only if $\frac{x^2 - 3x - 4}{6} = -1$. Rearranging gives $x^2 - 3x + 2 = (x - 1)(x - 2) = 0$, yielding possible solutions $x = 1$ and $x = 2$. $b(1) = -14$ is not a multiple of four, so $x = 1$ is not a solution; however, $b(2) = -8$, so $x = 2$ is a solution.

The solutions $x = -6, -2, 2, 3, 5$ sum to $\boxed{2}$.

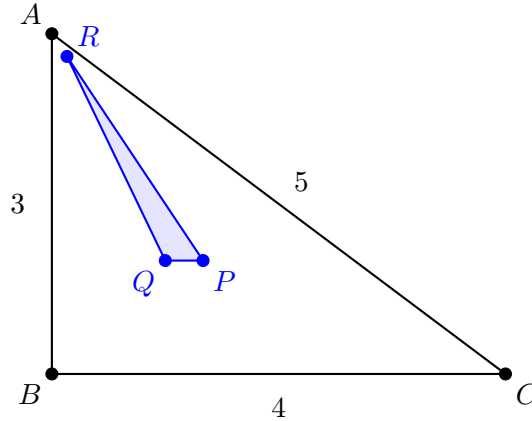
15. Let $[XYZ]$ denote the area of triangle $\triangle XYZ$. Right triangle $\triangle ABC$ has side lengths $AB = 3$, $BC = 4$, $AC = 5$. Points P , Q , and R are drawn inside $\triangle ABC$ such that:

- $[ABP] : [BCP] : [ACP] = 1 : 1 : 1$,
- $[ABQ] : [BCQ] : [ACQ] = 3 : 4 : 5$,
- $[ABR] : [BCR] : [ACR] = 1 : 28 : 1$.

What is the value of $[PQR]$?

Answer: $\frac{3}{10}$

Solution:



Recall that the medians of a triangle divide it into 6 triangles of equal area and concur at the centroid of the triangle. This matches the ratios given for P , and P is uniquely determined by these ratios, so P is the centroid of $\triangle ABC$.

Notice that

$$[ABQ] : [BCQ] : [ACQ] = 3 : 4 : 5 = AB : BC : AC.$$

This means that the heights of the three triangles from Q are the same length. Thus, Q is the incenter of $\triangle ABC$.

Let $A = (0, 3)$, $B = (0, 0)$, and $C = (4, 0)$. The coordinates of the centroid and incenter of $\triangle ABC$ are $P = (\frac{4}{3}, 1)$ and $Q = (1, 1)$.

The ratio $[ABR] : [BCR] : [ACR] = 1 : 28 : 1$ tells us that the area $[BCR] = \frac{28}{30} \cdot [ABC]$. Since these triangles share the same base, the height of $\triangle BCR$ is $\frac{28}{30} \cdot AB$.

Notice that $\overline{PQ}$ is parallel to $\overline{BC}$, so the height of $\triangle PQR$ from R to $\overline{PQ}$ has length $\frac{28}{30} \cdot 3 - 1$. Thus, the answer is

$$\begin{aligned} [PQR] &= \frac{1}{2} \cdot \left(\frac{4}{3} - 1 \right) \cdot \left(\frac{28}{30} \cdot 3 - 1 \right) \\ &= \frac{1}{2} \cdot \frac{1}{3} \cdot \frac{18}{10} \\ &= \boxed{\frac{3}{10}}. \end{aligned}$$

16. The Lunarians (moon people) define the *lunatic* operation, $\circledast$, on positive integers with the same number of digits. For any two positive integers $\underline{a_1} \dots \underline{a_n}$ and $\underline{b_1} \dots \underline{b_n}$, each with n digits,

$$\underline{a_1} \dots \underline{a_n} \circledast \underline{b_1} \dots \underline{b_n} = \underline{c_1} \dots \underline{c_n},$$

where for each $1 \leq i \leq n$, c_i is the units digit of the product $a_i b_i$. For example, $36 \circledast 82 = 42$, and $36 \circledast 820$ is not defined because 36 and 820 have different numbers of digits. For how many positive integers $a \leq 2025$ does there exist a positive integer $b \leq 2025$ for which $a \circledast b = 2025$?

Answer: 158

Solution:

To obtain 2025, the digits in each place value must combine under $\circledast$ to give 2, 0, 2, and 5 modulo 10, respectively. Consider addends in pairs, denoted $\underline{a_0 a_1 a_2 a_3}$ and $\underline{b_0 b_1 b_2 b_3}$. The leading

digits $\underline{a}_0$ and $\underline{b}_0$ must be 1 and 2 in some order, since any other combination would force one of the numbers to exceed 2025. Without loss of generality, let $\underline{a}_0 = 1$ and $\underline{b}_0 = 2$.

From here, $\underline{b}_1 = 0$ is necessary; any other choice of b_1 would make $\underline{b_0 b_1 b_2 b_3} > 2025$. With $\underline{b}_1 = 0$, $\underline{a}_1$ may take any value from 0 to 9, inclusive.

Next, b_2 must be 1 or 2: it cannot be 0 because the tens-digit product must end in 2, and it cannot exceed 2 because $b \leq 2025$. Also, b_3 must be odd because the units-digit product must end in 5. Thus there are eight possible values of $\underline{b_0 b_1 b_2 b_3}$ beginning with $\underline{20}$: the eight odd numbers from 2011 through 2025, inclusive.

If $b_2 = 1$, then $a_2 = 2$; if $b_2 = 2$, then a_2 is 1 or 6. Hence $a_2 \in \{1, 2, 6\}$, and a_3 must be odd. Any choice of digits $\underline{a}_2 \in \{1, 2, 6\}$ and $\underline{a}_3 \in \{1, 3, 5, 7, 9\}$ yields 2025 when combined under $\otimes$ with at least one of 2015 or 2025. Therefore all 15 possible values of $\underline{a_2 a_3}$ admit a choice of $\underline{b_0 b_1 b_2 b_3}$ such that

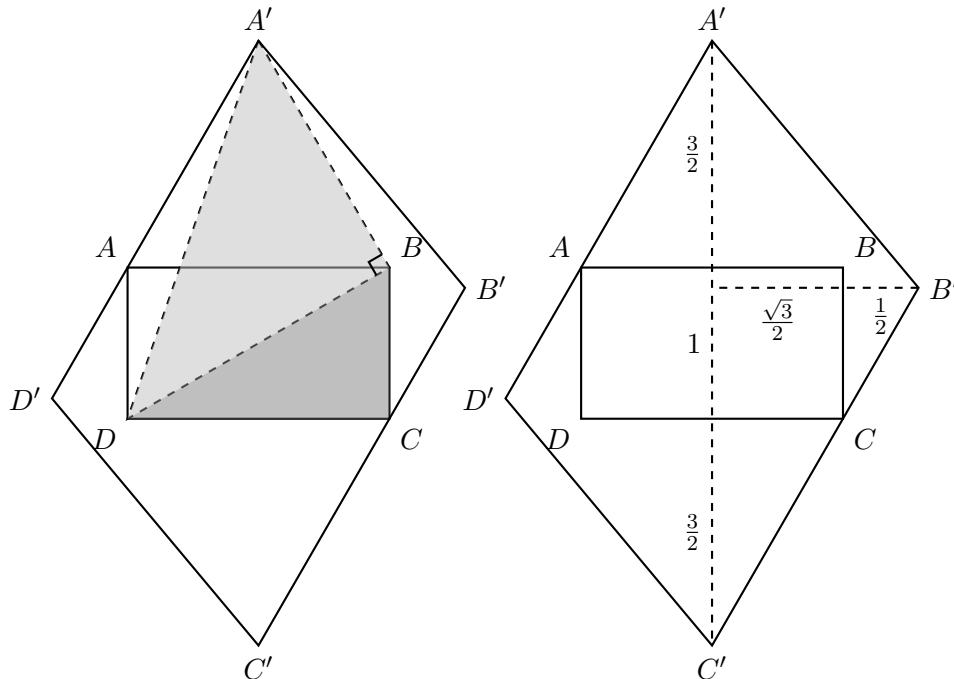
$$\underline{a_0 a_1 a_2 a_3} \otimes \underline{b_0 b_1 b_2 b_3} = 2025.$$

Since there are 10 choices for $\underline{a}_1$, this gives $10 \cdot 15 = 150$ possible values of a beginning with 1. In addition, there are the 8 values of a beginning with 20 described above. Thus, the answer is $8 + 150 = \boxed{158}$.

17. Rectangle $ABCD$ has $AB = CD = \sqrt{3}$ and $AD = BC = 1$. Points A', B', C', D' are constructed such that $AB = A'B, BC = B'C, CD = C'D, AD = AD'$ and the areas of quadrilaterals $A'BCD$, $AB'CD$, $ABC'D$, and $ABCD'$ are all maximized. Find the area of quadrilateral $A'B'C'D'$.

Answer: $2 + 2\sqrt{3}$ or $\frac{4}{\sqrt{3}-1}$

Solution:



The essence of this problem is determining how to maximize the area of one of the quadrilaterals. The strategy is the same for all quadrilaterals, so use $A'BCD$ as an example. No matter where A'

is located, the area of $\triangle BCD$ is fixed, so we are just maximizing the area of $\triangle A'BD$. Since the base BD is fixed, we are looking for the maximum possible height given by point A' , which lives somewhere on a circle of radius AB centered at B . This is achieved exactly when $A'B \perp BD$. B', C', D' can be found similarly.

To find the area of $A'B'C'D'$, use $30-60-90$ triangles to see that ABA' is equilateral, so the length of diagonal $A'C'$ is $\frac{\sqrt{3}}{2}AB + 1 + \frac{\sqrt{3}}{2}CD = 4$.

The height of triangle $\triangle A'B'C'$ from the base $\overline{A'C'}$ is $\frac{1}{2}AB + \frac{1}{2}B'C = \frac{1+\sqrt{3}}{2}$.

This gives a total area of $2 \cdot \frac{1}{2} \cdot 4 \cdot \frac{1+\sqrt{3}}{2} = \boxed{2 + 2\sqrt{3}}$.

18. A base-2025 palindrome is a positive integer that reads the same forward and backward when written in base-2025. For each positive integer n , let $P(n)$ be the number of integers between 1 and n , inclusive, that are base-2025 palindromes. Find the smallest real number x such that that $x\sqrt{n} \geq P(n)$ for all positive integers n .

Answer: $\frac{2026}{45}$

Solution: Let $b = 2025$. We claim the least constant is

$$C = \frac{b+1}{\sqrt{b}} = \frac{2026}{45}.$$

A d -digit base- b palindrome is determined by its first $\lceil d/2 \rceil$ digits, so there are $(b-1)b^{\lceil d/2 \rceil - 1}$ such palindromes. Hence there are $(b-1)b^{j-1}$ palindromes with $2j-1$ digits and the same number with $2j$ digits.

First, we find a lower bound on C . Since b^{2k-1} is not a palindrome, $P(b^{2k-1})$ counts all palindromes with at most $2k-1$ digits. Pairing the $(2j-1)$ - and $2j$ -digit palindromes for $j = 1, \dots, k-1$, and adding the $(2k-1)$ -digit ones,

$$P(b^{2k-1}) = 2(b-1)(1 + b + \dots + b^{k-2}) + (b-1)b^{k-1} = (b+1)b^{k-1} - 2.$$

Thus

$$\frac{P(b^{2k-1})}{\sqrt{b^{2k-1}}} = \frac{b+1}{\sqrt{b}} - \frac{2}{b^{k-1/2}} \rightarrow \frac{b+1}{\sqrt{b}}$$

as $k \rightarrow \infty$. Therefore, any valid C must satisfy $C \geq (b+1)/\sqrt{b}$.

Next, we show that this value actually works. For $n < b$, we have $P(n) = n$, so

$$\frac{P(n)}{\sqrt{n}} = \sqrt{n} < \sqrt{b} < \frac{b+1}{\sqrt{b}}.$$

Now take $n \geq b$ and choose $k \geq 1$ such that

$$b^{2k-1} \leq n < b^{2k+1}.$$

Any palindrome in this range has $2k$ or $2k+1$ digits. In either case, it is uniquely determined by all digits preceding its final k digits.

Partition the integers into blocks

$$[mb^k, (m+1)b^k).$$

Within each block, all digits preceding the final k digits are fixed, so each block contains at most one palindrome. The first relevant block has index b^{k-1} , and n lies in block $\lfloor n/b^k \rfloor$. Hence

$$P(n) \leq P(b^{2k-1}) + \left\lfloor \frac{n}{b^k} \right\rfloor - b^{k-1} + 1 \leq b^k + \frac{n}{b^k} - 1 < b^k + \frac{n}{b^k}.$$

It remains to compare $b^k + \frac{n}{b^k}$ with $\frac{b+1}{\sqrt{b}}\sqrt{n}$ on $[b^{2k-1}, b^{2k+1}]$. The two expressions agree at the endpoints, and $b^k + \frac{n}{b^k}$ is linear in n while $\frac{b+1}{\sqrt{b}}\sqrt{n}$ is concave. Since a concave curve lies above its chords,

$$b^k + \frac{n}{b^k} \leq \frac{b+1}{\sqrt{b}}\sqrt{n}.$$

Therefore

$$P(n) < \frac{b+1}{\sqrt{b}}\sqrt{n}$$

for all n , and the answer is

$$C = \frac{b+1}{\sqrt{b}} = \boxed{\frac{2026}{45}}.$$

19. Let N_{21} be the answer to problem 21.

Four distinct points in the coordinate plane are chosen uniformly at random from the set

$$\{(21, N_{21}), (-21, N_{21}), (21, -N_{21}), (-21, -N_{21}), (N_{21}, 21), (-N_{21}, 21), (N_{21}, -21), (-N_{21}, -21)\}.$$

Given that there is one point in each quadrant, what is the expected area of the convex quadrilateral with vertices at the four chosen points?

Answer: 1936

Solution: For each shape that could possibly be made, we will calculate its area and its probability of appearing. Let $A = 21$. Starting from the tall rectangle configuration $(\pm A, \pm N_{21})$, each quadrant's point can "flip" to $(\pm N_{21}, \pm A)$ with probability $\frac{1}{2}$. Different shapes appear based on the number of flips that occur.

Zero and four flips yield rectangles with area $2N_{21} \cdot 2A$.

With one and three flips, one gets an asymmetric shape whose area can be thought of as a rectangle of side lengths $N_{21} + A, 2A$ and two triangles each of base lengths $N_{21} - A, N_{21} + A$. This total area is $2A \cdot (N_{21} + A) + (N_{21} + A)(N_{21} - A) = (N_{21} + A)^2$.

With two flips, two types of shapes occur. The trapezoid has a long side with length $2N_{21}$, a short side with length $2A$, and a height of $N_{21} + A$, so its area is $\frac{1}{2}(2N_{21} + 2A)(N_{21} + A) = (N_{21} + A)^2$. The tilted square has a side length s which is the hypotenuse of a right triangle with base lengths

$N_{21} + A$ and $N_{21} - A$, so by Pythagoras' theorem the area is $s^2 = (N_{21} + A)^2 + (N_{21} - A)^2 = 2(N_{21}^2 + A^2)$.

Given that two flips occurred, the probability of the shape being a trapezoid is $2/3$, and the probability of it being a tilted square is $1/3$. One way to see this is that there are four possible trapezoid configurations because there are four ways that the short side can be oriented; but there are only two tilted squares, the left-tilted one and the right-tilted one.

Now, let's calculate the probability of each case. The total number of ways to choose four points is $2^4 = 16$, so the probability for zero or four flips is $\frac{1}{16}(\binom{4}{0} + \binom{4}{4}) = \frac{2}{16}$, for one or three flips, $\frac{1}{16}(\binom{4}{1} + \binom{4}{3}) = \frac{8}{16}$, and for two flips, $\frac{1}{16}\binom{4}{2} = \frac{6}{16}$. This means that the probability for a trapezoid is $\frac{4}{16}$ and the probability for a tilted square is $\frac{2}{16}$. Then the expected area is $\frac{2}{16} \cdot 2N_{21} \cdot 2A + \frac{8}{16}(N_{21} + A)^2 + \frac{4}{16}(N_{21} + A)^2 + \frac{2}{16}2(N_{21}^2 + A^2) = \frac{1}{4}(2N_{21}A + 2(N_{21} + A)^2 + (N_{21} + A)^2 + (N_{21}^2 + A^2)) = \frac{4}{4}(N_{21} + A)^2$. Plugging in $N_{21} = 23$, $A = 21$ gives $44^2 = \boxed{1936}$.

20. Let N_{19} be the **leading two digits of the decimal representation** of the answer to problem 19.

The 24 lines $y = \pm 4x, y = \pm 8x, \dots, y = \pm 48x$ and the curve $y = N_{19}x^2 + N_{19}$ are drawn in the coordinate plane. How many (possibly infinitely large) distinct regions of the coordinate plane are in the resulting graph?

Answer: 61

Solution: Write the quadratic as $y = ax^2 + bx + c$, where $a = c = N_{19} = 19$ and $b = 0$. We can classify the lines $y = kx$ by whether they intersect the quadratic zero, one, or two times; each type of line contributes a different number of regions. Since solving for an intersection gives a quadratic equation, no line can intersect the parabola more than twice.

Suppose there are a_0 lines that never intersect the quadratic, a_1 lines that intersect it once, and a_2 lines that intersect it twice. We first determine the number of regions in terms of a_0, a_1, a_2 .

Start with just the quadratic and the line $y = 0$, which create three regions. Since all of the lines $y = kx$ meet at the origin, adding a line that never intersects the quadratic contributes two new regions; adding a tangent line contributes three; and adding a line that intersects the quadratic twice contributes four. Since $y = 0$ belongs to the a_0 class, the total number of regions is

$$R = 3 + 2(a_0 - 1) + 3a_1 + 4a_2 = 1 + 2a_0 + 3a_1 + 4a_2.$$

Tangency occurs when the equation

$$kx = ax^2 + bx + c$$

has exactly one solution. This happens when its discriminant is zero:

$$(b - k)^2 - 4ac = 0.$$

Since $a = c$ and $b = 0$, the tangent slopes are $k_{\pm} = \pm 2a = \pm 38$.

A line $y = kx$ does not intersect the quadratic when $k_- < k < k_+$, is tangent when $k = k_{\pm}$, and intersects it twice otherwise.

We can now classify the given lines. Since $a = 19$, the lines with slopes $\pm 4, \pm 8, \dots, \pm 36$ belong to the a_0 class, so $a_0 = 18$. There are no lines with slopes ± 38 , so $a_1 = 0$. Since there are 24 lines in total,

$$a_2 = 24 - 18 = 6.$$

Therefore,

$$R = 1 + 2 \cdot 18 + 3 \cdot 0 + 4 \cdot 6 = \boxed{61}.$$

21. Let N_{20} be the answer to problem 20.

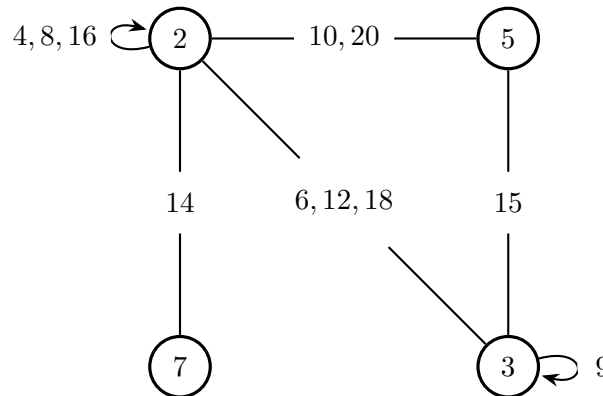
Clara needs to visit $N = \lfloor \frac{N_{20}}{3} \rfloor$ places which are labeled with the integers $0, 2, 3, 4, \dots, N$ (there is no place 1). Each pair of places a and b is connected if and only if $\gcd(a, b) > 1$ (note that $\gcd(0, n) = n$ for all positive integers n). Call every instance when Clara moves from one place to a connected place a *trip*. If Clara starts at place 0, what is the minimum number of trips she needs to make to visit all places?

Answer: 23

Solution: The answer to problem 20 is 61, so $N = \lfloor \frac{N_{20}}{3} \rfloor = \lfloor \frac{61}{3} \rfloor = 20$.

All places numbered with primes greater than $\frac{20}{2} = 10$ are connected only to place 0. Since Clara starts at 0, she must return to 0 at least four times to visit all these components. She must also visit each of the 19 nonzero places, so at least $19 + 4 = 23$ trips are necessary.

If Clara visits the four primes greater than 10 first and returns to 0 after each visit, these visits use 8 trips. To determine the most efficient way to visit the remaining places, we can represent them by a graph with the primes less than 10 as vertices and label edges with composite numbers divisible by both primes. Loops are labeled with powers of the corresponding prime:



This representation helps us find a path that visits each remaining place exactly once. For example,

$$0 \rightarrow 7 \rightarrow 14 \rightarrow 2 \rightarrow 4 \rightarrow 8 \rightarrow 16 \rightarrow 6 \rightarrow 12 \rightarrow 18 \rightarrow 3 \rightarrow 9 \rightarrow 15 \rightarrow 5 \rightarrow 20 \rightarrow 10$$

is one such path, and it uses 15 trips. Thus, the minimum number of trips is achieved: $8 + 15 = \boxed{23}$.

Once we have solved problems 19, 20, and 21 in terms of the variables N_{19}, N_{20}, N_{21} , we can use the relationships between the answers to solve for all three answers simultaneously. Recall from problem 20 that a_i is the number of lines meeting the parabola at exactly i points, and the number of regions is

$$N_{20} = 1 + 2a_0 + 3a_1 + 4a_2.$$

A line with slope m belongs to the a_0 , a_1 , or a_2 class according as $|m| < 2N_{19}$, $|m| = 2N_{19}$, or $|m| > 2N_{19}$. Thus, the number of regions is nonincreasing as N_{19} increases.

First, the answer to problem 20 is between $1 + 2 \cdot 24$ and $1 + 4 \cdot 24$, so $16 \leq N \leq 32$. The answer to problem 21 is a nondecreasing function of N : any route for a larger value of N gives a route for a smaller value by replacing stops above the smaller bound with 0 and removing consecutive repetitions of 0. Thus, the answers for $N = 16$ and $N = 32$ bound N_{21} .

For $N = 16$, there are two primes greater than 8, so the same lower-bound argument gives at least $16 - 1 + 2 = 17$ trips. This is attained by the route

$$0 \rightarrow 7 \rightarrow 14 \rightarrow 2 \rightarrow 6 \rightarrow 3 \rightarrow 9 \rightarrow 12 \rightarrow 8 \rightarrow 16 \rightarrow 4 \rightarrow 10 \rightarrow 5 \rightarrow 15 \rightarrow 0 \rightarrow 13 \rightarrow 0 \rightarrow 11.$$

For $N = 32$, there are five primes greater than 16, so the lower bound is $32 - 1 + 5 = 36$ trips. This is attained by visiting these five primes first, returning to 0 after each visit, and then following

$$\begin{aligned} 0 &\rightarrow 11 \rightarrow 22 \rightarrow 2 \rightarrow 4 \rightarrow 8 \rightarrow 16 \rightarrow 32 \rightarrow 24 \rightarrow 12 \rightarrow 6 \rightarrow 18 \rightarrow 9 \rightarrow 27 \\ &\rightarrow 3 \rightarrow 15 \rightarrow 5 \rightarrow 25 \rightarrow 10 \rightarrow 20 \rightarrow 30 \rightarrow 21 \rightarrow 7 \rightarrow 14 \rightarrow 28 \rightarrow 26 \rightarrow 13. \end{aligned}$$

Thus, $17 \leq N_{21} \leq 36$. The answer to problem 19 is therefore between $(17 + 21)^2 = 1444$ and $(36 + 21)^2 = 3249$, so its first two digits, N_{19} , must be between 14 and 32.

Working through problem 20 with this in mind, we can tighten our bounds. For $N_{19} = 14$, the lines with slopes $\pm 4, \dots, \pm 24$ give $a_0 = 12$, the lines with slopes ± 28 give $a_1 = 2$, and $a_2 = 24 - 12 - 2 = 10$, so the number of regions is 71. For $N_{19} = 32$, all lines belong to the a_0 class because $2 \cdot 32 = 64 > 48$, so the number of regions is 49. Thus, $16 \leq N \leq 23$.

For every integer N in this interval, let k be the number of primes greater than $N/2$ and at most N . The same lower-bound argument gives at least $N - 1 + k$ trips. This bound is attained by visiting these k primes first, returning to 0 after each visit, and then following the original path for $N = 20$ with the following modifications: insert 21 between 3 and 9, append $22 \rightarrow 11$, and delete all labels greater than N and all primes greater than $N/2$. The resulting path visits every remaining place exactly once, and consecutive labels have a common factor greater than 1. Hence, the minimum is $N - 1 + k$ throughout this interval.

For $N = 23$, there are four primes greater than $23/2$, so the answer is $23 - 1 + 4 = 26$. Thus, the answer to problem 19 is between 1444 and $(26 + 21)^2 = 2209$, giving $14 \leq N_{19} \leq 22$.

Tightening our bounds again, for $N_{19} = 22$, the lines with slopes $\pm 4, \dots, \pm 40$ give $a_0 = 20$, the lines with slopes ± 44 give $a_1 = 2$, and $a_2 = 24 - 20 - 2 = 2$, so the number of regions is 55. Thus, $18 \leq N \leq 23$. For $N = 18$, there are three primes greater than 9, so the answer is $18 - 1 + 3 = 20$. The answer to problem 19 is therefore between $(20 + 21)^2 = 1681$ and 2209, giving $16 \leq N_{19} \leq 22$.

Tightening our bounds again, for $N_{19} = 16$, the lines with slopes $\pm 4, \dots, \pm 28$ give $a_0 = 14$, the lines with slopes ± 32 give $a_1 = 2$, and $a_2 = 24 - 14 - 2 = 8$, so the number of regions is 67. Thus, $18 \leq N \leq 22$. For $N = 22$, there are three primes greater than 11, so the answer is

$22 - 1 + 3 = 24$. The answer to problem 19 is therefore between 1681 and $(24 + 21)^2 = 2025$, giving $16 \leq N_{19} \leq 20$.

For $N_{19} = 20$, the lines with slopes $\pm 4, \dots, \pm 36$ give $a_0 = 18$, the lines with slopes ± 40 give $a_1 = 2$, and $a_2 = 24 - 18 - 2 = 4$, so the number of regions is 59. Thus, $19 \leq N \leq 22$. For $N = 19$, there are four primes greater than $19/2$, so the answer is $19 - 1 + 4 = 22$. The answer to problem 19 is therefore between $(22 + 21)^2 = 1849$ and 2025, giving $18 \leq N_{19} \leq 20$.

Finally, for $N_{19} = 18$, the lines with slopes $\pm 4, \dots, \pm 32$ give $a_0 = 16$, the lines with slopes ± 36 give $a_1 = 2$, and $a_2 = 24 - 16 - 2 = 6$, so the number of regions is 63. Thus, $19 \leq N \leq 21$.

For $N = 19$, we have $N_{21} = 22$, so the answer to problem 19 is 1849, giving $N_{19} = 18$. The answer to problem 20 is then 63, as established. This implies $N = \lfloor 63/3 \rfloor = 21$, which is inconsistent.

For $N = 21$, there are four primes greater than $21/2$, so $N_{21} = 21 - 1 + 4 = 24$. The answer to problem 19 is then 2025, giving $N_{19} = 20$. The answer to problem 20 is 59, as established. This implies $N = \lfloor 59/3 \rfloor = 19$, which is inconsistent.

For $N = 20$, there are four primes greater than $20/2$, so $N_{21} = 20 - 1 + 4 = 23$. The answer to problem 19 is then 1936, giving $N_{19} = 19$. In problem 20, the lines with slopes $\pm 4, \dots, \pm 36$ give $a_0 = 18$, while $a_1 = 0$ and $a_2 = 24 - 18 = 6$, so the answer is 61. This implies $N = \lfloor 61/3 \rfloor = 20$, which is consistent. Therefore, the answers 1936, 61, 23 are correct and unique.

22. Aarush and Brian are watching a sequence of fair coin flips. Both players start with 0 points. Aarush gains a point when the coin lands on heads, and Brian gains a point when the coin lands on tails. If Brian reaches 2 points, both players reset their points to 0. What is the expected number coin flips it takes for Aarush to reach 9 points for the first time?

Answer: 370

Solution: Call the flips from a reset until either nine heads or two tails appear a round. Let p be the probability that a round ends with nine heads, and let L be the number of flips in a round.

A round lasts at most ten flips. Among the 2^{10} equally likely ten-flip strings, success occurs exactly for the all-heads string or a string with one tail. (Flips after the round ends may be ignored.) Therefore,

$$p = \frac{1 + 10}{2^{10}} = \frac{11}{1024}.$$

For $0 \leq j \leq 8$, the round is still running after j flips exactly when at most one tail has appeared. There are $1 + j$ such j -flip strings, so $\Pr(L > j) = (j + 1)/2^j$. After nine flips, the round is still running exactly when one tail has appeared, so $\Pr(L > 9) = 9/2^9$. Thus,

$$\mathbb{E}[L] = \sum_{j=0}^9 \Pr(L > j) = \sum_{j=0}^8 \frac{j+1}{2^j} + \frac{9}{2^9} = \left(4 - \frac{11}{256}\right) + \frac{9}{512} = \frac{2035}{512}.$$

Let T be the expected total number of flips. After an unsuccessful round, the process restarts from the same state, giving

$$T = \mathbb{E}[L] + (1 - p)T.$$

Finally,

$$T = \frac{\mathbb{E}[L]}{p} = \frac{2035}{512} \cdot \frac{1024}{11} = \boxed{370}.$$

23. There is a unique finite sequence of positive integers $(a_1, \dots, a_n)$ such that $a_i^2 \leq a_{i+1}$ for each $1 \leq i < n$ and

$$\prod_{j=1}^n \left(1 + \frac{1}{a_j}\right) = 2025.$$

Find $n + a_n$.

Answer: 526.

Solution: The statement guarantees uniqueness, so it suffices to construct a sequence satisfying both conditions. We employ a greedy strategy, appending the smallest positive integer that satisfies the growth condition and keeps the partial product at most 2025. Start with ten 1's, whose factors contribute $2^{10} = 1024$. Appending 2 and 4 brings the product to

$$1024 \cdot \frac{3}{2} \cdot \frac{5}{4} = 1920.$$

The remaining factor needed is $2025/1920 = 135/128$. Appending 19 contributes $20/19$, leaving

$$\frac{135}{128} \cdot \frac{19}{20} = \frac{513}{512},$$

which is exactly the factor contributed by appending 512.

Thus consider

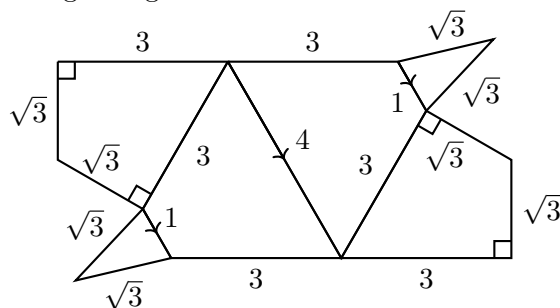
$$\underbrace{(1, \dots, 1)}_{10 \text{ terms}}, 2, 4, 19, 512).$$

The growth condition holds because $1^2 \leq 1$, $1^2 \leq 2$, $2^2 = 4$, $4^2 \leq 19$, and $19^2 \leq 512$. The requested product is achieved:

$$2^{10} \cdot \frac{3}{2} \cdot \frac{5}{4} \cdot \frac{20}{19} \cdot \frac{513}{512} = 2025.$$

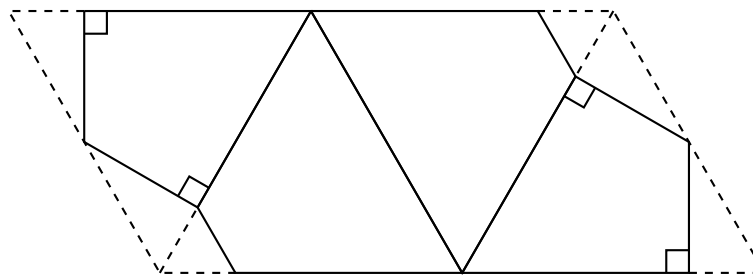
We have found the desired sequence, and the answer is $n + a_n = 14 + 512 = \boxed{526}$.

24. Determine the volume of the three-dimensional solid with the net shown below. Arrows denote parallel segments, and some right angles are marked.

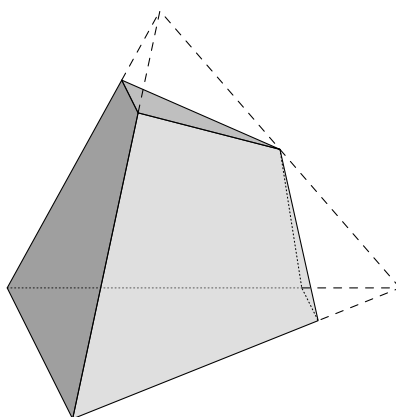


Answer: $5\sqrt{2}$ or $\frac{10}{\sqrt{2}}$

Solution: From the side lengths of the two kites, we see that the angles are 120° and 60° . Using the parallel lines, this allows us to find that all angles in the diagram, except for the triangles, are 120° or 60° .



In particular, the two central vertices are equivalent to that of a regular tetrahedron. So, if we extend the horizontal sides to be length 4 instead of 3, the resulting net is exactly like a tetrahedron of side length 4.

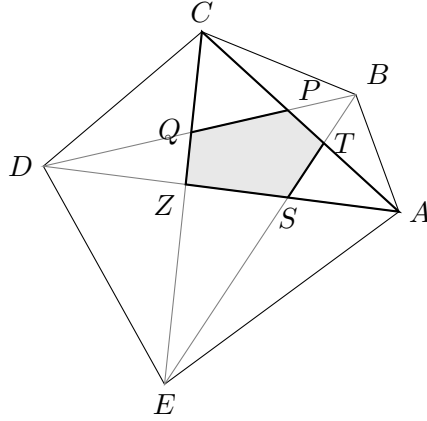


So, the shape we have is just a doubly truncated tetrahedron. From symmetry, the vertex where all the side of length $\sqrt{3}$ meet up is the midpoint of the auxillary side of length 4. From this, we see that the two smaller tetrahedra that are removed from the large tetrahedron have $\frac{1}{16}$ the base and $\frac{1}{2}$ the height of our large tetrahedron and thus have $\frac{1}{32}$ the volume. Using any method, the volume of the large tetrahedron can be calculated to be $V = \frac{64}{6\sqrt{2}}$, and removing the 2 truncations gives $\frac{30}{32}V = \boxed{5\sqrt{2}}$.

25. Pentagon $ABCDE$ has $AB = 3, BC = 4, CD = 5, DE = 6$, and $EA = 7$, with all vertices lying on a circle. The five diagonals of $ABCDE$ enclose a smaller pentagon. Compute the area of this smaller pentagon. Submit your answer as a positive real number E to at most 5 decimal places; if the correct answer is A , your score for this problem is round $\left(25 \exp \left(-\frac{3}{2} \left(\max \left(\frac{E^2}{A^2}, \frac{A}{E}\right) - 1\right)^{\frac{2}{3}}\right)\right)$.

Answer: 4.9072848

Solution:



Let P, Q, Z, S, T be the intersections of the diagonals of $ABCDE$, as shown in the diagram above. A useful way to calculate the area of the interior pentagon $PQZST$ is

$$[PQZST] = [ACZ] - [AST] - [CPQ].$$

We first estimate the diameter d of the circle. For $a = 3, 4, 5, 6, 7$, let $u_a = \sqrt{d^2 - a^2}$, and let h_a be half the central angle subtended by the side of length a . Then

$$\sin h_a = \frac{a}{d}, \quad \cos h_a = \frac{u_a}{d}, \quad h_3 + h_4 + h_5 + h_6 + h_7 = 180^\circ.$$

Consequently, $\cos(h_3 + h_4 + h_5) + \cos(h_6 + h_7) = 0$. Expanding with cosine sum identities and multiplying by d^3 gives

$$u_5(u_3u_4 - 12) - 5(3u_4 + 4u_3) + d(u_6u_7 - 42) = F(d) = 0.$$

The circumcircle of a regular pentagon with side 5 has diameter about 8.5, which motivates testing nearby values:

d	u_3	u_4	u_5	u_6	u_7	$F(d)$
8.5	7.95	7.50	6.87	6.02	4.82	-54.68
8.6	8.06	7.61	7.00	6.16	5.00	-26.31
8.7	8.17	7.73	7.12	6.30	5.17	2.83

$F(8.7)$ is very close to zero, so we use $d \approx 8.7$.

Triangles sharing an angle have areas in the ratio of the products of the two adjacent sides. Our desired area is thus

$$[ACD] \left(\frac{AZ}{AD} - \frac{AS}{AD} \frac{AT}{AC} \right) - [ACE] \frac{CP}{CA} \frac{CQ}{CE}$$

so it is helpful to calculate the length ratios that appear in this expression.

Let $p = AC$, $q = BD$, $r = CE$, $s = AD$, and $t = BE$. The chord joining the endpoints of two consecutive sides a, b has length

$$d \sin(h_a + h_b) = \frac{au_b + bu_a}{d}.$$

Applying this to the five consecutive pairs gives

$$(p, q, r, s, t) \approx (6.42, 7.72, 8.53, 8.63, 8.36).$$

For instance, $AC \approx (3(7.73) + 4(8.17))/8.7 \approx 6.42$.

In a cyclic quadrilateral $WXYZ$ whose diagonals intersect at J , similar triangles give

$$\frac{WJ}{JY} = \frac{WX \cdot WZ}{YX \cdot YZ}.$$

Using this identity at the five intersections gives

ratio	expression	estimate
AZ/AD	$7p/(30 + 7p)$	0.600
AS/AD	$21/(21 + 6q)$	0.312
AT/AC	$21/(21 + 4r)$	0.381
CP/CA	$20/(20 + 3s)$	0.436
CQ/CE	$20/(20 + 6t)$	0.285

The area of a triangle inscribed in a circle of diameter d is the product of its sides divided by $2d$. Therefore

$$[ACD] = \frac{5ps}{2d} \approx 15.92, \quad [ACE] = \frac{7pr}{2d} \approx 22.03.$$

Our final estimate is

$$[ACD] \left(\frac{AZ}{AD} - \frac{AS}{AD} \frac{AT}{AC} \right) - [ACE] \frac{CP}{CA} \frac{CQ}{CE} \approx 15.92(0.600 - 0.312 \cdot 0.381) - 22.03(0.436)(0.285) \approx 4.92$$

which earns 24 points.

26. A polynomial is called *short* if its coefficients are all integers between -5 and 5 , inclusive. Compute the number of short polynomials of degree at most 5 which can be written as the product of two nonconstant short polynomials. Submit your answer as a positive integer E ; if the correct answer is A , your score for this problem is $\text{round} \left(25 \exp \left(\frac{1}{3} \left(1 - \max \left(\frac{E^3}{A^3}, \frac{A}{E} \right)^3 \right) \right) \right)$.

Answer: 307462

Solution: If we're really short on time, we should probably guess a number between 0 and the number of short polynomials with degree at most 5. There are 11 choices for each coefficient in a short polynomial, and if the polynomial has degree at most 5, there are 6 coefficients to make this choice for. Thus, a very loose upper bound on the answer here is $11^6 = 1771561$. Also, a loose lower bound is $11^5 - 11 = 161040$ since setting the constant term to zero immediately gives the product of a short polynomial and x (excluding 11 constant quotients). The score function penalizes overestimates a lot more than underestimates, so it's probably best to guess closer to the lower bound.

We can get a better estimate by counting products with one of the factors $x - 1$, or $x + 1$. Multiplication by $x \pm 1$ adds or subtracts adjacent coefficients, so these families are relatively easy to count. In each family we require both the product and the quotient to be short.

For $x - 1$, write the quotient as $b_0 + b_1x + \dots + b_4x^4$, allowing leading zeros. The product is short if and only if consecutive coefficients differ by at most 5. Let $w_k(v)$ count the number of strings of k coefficients in $[-5, 5]$ satisfying this condition and ending at v . We have $w_k(-v) = w_k(v)$. The following table can be constructed recursively via

$$w_{k+1}(v) = \sum_{u=\max(-5, v-5)}^{\min(5, v+5)} w_k(u)$$

Note the total column is the sum from $v = -5$ to $v = 5$:

$k \backslash v$	0	1	2	3	4	5	total
1	1	1	1	1	1	1	11
2	11	10	9	8	7	6	91
3	91	85	78	70	61	51	781
4	781	730	669	599	521	436	6691
5	6691	6255	5734	5135	4466	3736	57343

Excluding constant quotients gives $57343 - 11 = 57332$ products. Substituting $-x$ for x gives the same count for the factor $x + 1$.

We now account for products counted twice. A product in the $x - 1$ family is also in the x family exactly when $b_0 = 0$. The remaining four coefficients form an allowed string, so this overlap has size $6691 - 1 = 6690$, excluding the zero polynomial. The overlap for x and $x + 1$ is the same.

For the remaining overlap, an upper bound will suffice. If a polynomial belongs to both the $x - 1$ and $x + 1$ families, write it as

$$P = (x^2 - 1)(a + bx + cx^2 + dx^3).$$

Its two short quotients contain the coefficients $a + b, a - b$ and $c + d, c - d$. Consequently

$$|a| + |b| \leq 5, \quad |c| + |d| \leq 5.$$

There are $11 + 2(9 + 7 + 5 + 3 + 1) = 61$ integer pairs satisfying either inequality. Ignoring the other coefficient restrictions therefore gives at most $61^2 - 1 = 3720$ products in this overlap, excluding zero.

Subtracting all three pairwise overlaps gives a lower bound; products in all three families are subtracted too often, which only makes this bound smaller. Hence there are at least

$$161040 + 2(57332) - 2(6690) - 3720 = 258604$$

desired polynomials; submitting 258604 is worth 20 points. We have certainly undercounted and could probably squeeze a few more points out of this solution under contest conditions by submitting a slightly higher number.

27. The Euclidean algorithm allows one to find the greatest common divisor of two integers by repeatedly applying the fact that, if $a \leq b$, $\gcd(a, b) = \gcd(b \bmod a, a)$, where $b \bmod a$ is the remainder after dividing b by a . This algorithm continuously replaces b with $b \bmod a$ until one of the numbers is zero. For example, we can find the gcd of 10 and 12 as follows: $\gcd(10, 12) = \gcd(2, 10)$ because $12 \bmod 10$ is 2, which equals $\gcd(0, 2)$ because $10 \bmod 2$ is 0. The other term, 2, is thus $\gcd(10, 12)$.

Let $f(a, b)$ be the **number of iterations** in the Euclidean algorithm. For example, $f(10, 12) = 2$, $f(2, 4) = 1$, and $f(5, 8) = 4$. Find $f(1, 2025) + f(2, 2025) + \cdots + f(2025, 2025)$. Submit your answer as a positive integer E ; if the correct answer is A , your score for this problem is $\text{round}\left(25 \exp\left(-4.5 \left(\frac{|E-A|}{A}\right)\right)\right)$.

Answer: 12337

Solution: For this problem, taking a random sample is probably faster and less error-prone than trying to find a clever solution. If we manually calculate $f(a, 2025)$ for a few random integers a

between 1 and 2025 and guess that our sample mean is the population mean, multiplying by 2025 gives an unbiased estimate of the true answer. This strategy allows teammates to parallelize the work, so this isn't as tedious as it sounds!

Here is one way to generate random numbers between 1 and 2025 in your head. Generate a stream of random 1's and 0's (for example, considering the parity of the digits of π yields a stream of 1's and 0's, and it is reasonable to assume this stream is unbiased, even though there's nothing random about it!) and interpret the result as a stream of 11-digit binary numbers. Then throw out any 0's or numbers greater than 2025. Try thinking of your own scheme, too!

We may like to know how the number of points we expect to get scales with the number of samples we take, e.g., to know how many points an extra computation is worth. This is hard to estimate under contest conditions, but simulating this strategy with code shows that 6 samples (one per teammate!) is good enough for about 16 points on average.